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Project Report

Project Titled Name

**Simulation-Based Performance and Energy Efficiency Analysis of a Centrifugal
Pump Under Varying Operating Conditions Using Aspen HYSYS**

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Abstract

This project presents a simulation-based performance and sensitivity analysis of a centrifugal pump using Aspen HYSYS 11. Water was selected as the working fluid and a base pump case was developed with a flow rate of 10,000 kg/h, inlet pressure of 1 bar, outlet pressure of 5 bar, inlet temperature of 30 °C, and specified adiabatic efficiency of 70%. Four one-variable-at-a-time studies were then performed to investigate the effects of fluid temperature, flow rate, discharge pressure, and pump efficiency. A total of 21 operating cases were evaluated. Pump head, power consumption, pressure rise, outlet temperature, and efficiency were recorded. Specific energy consumption was also calculated to provide an energy-performance indicator. The results show that flow rate and discharge pressure have strong effects on pump power, temperature has a comparatively small effect over the investigated range, and higher pump efficiency substantially reduces required power and specific energy consumption.

Keywords

Centrifugal pump; Aspen HYSYS 11; pump efficiency; pump head; pressure rise; power consumption; flow rate; temperature; specific energy consumption; sensitivity analysis.

1. Introduction

Pumping systems are widely used in chemical and process industries to transport liquids between process units, storage vessels, heat-transfer equipment, and other operations. The energy consumed by pumps depends on the required flow rate, pressure rise, fluid properties, and pump efficiency [2,3]. Understanding these relationships is important for equipment operation, energy management, and preliminary process design.

In this project, Aspen HYSYS 11 was used to model a centrifugal pump handling water [1]. Rather than considering only a single operating point, the model was subjected to controlled sensitivity studies. The purpose was to quantify how changes in temperature, flow rate, discharge pressure, and specified pump efficiency affect the calculated pump performance.

2. Aim and Objectives

The main aim of the project was to evaluate centrifugal pump performance under varying operating conditions using Aspen HYSYS 11.

- Develop a steady-state water pumping model in Aspen HYSYS 11.
- Establish a base operating case for comparison.
- Investigate the effect of fluid temperature on pump performance.
- Investigate the effect of flow rate on pump head and power consumption.
- Investigate the effect of discharge pressure on pump head and power consumption.
- Investigate the effect of specified pump adiabatic efficiency on required power.
- Calculate specific energy consumption as an additional energy-performance indicator.
- Interpret the trends and develop practical engineering recommendations

3. Software and Simulation Basis

Item	Specification
Process simulator	Aspen HYSYS 11
Working fluid	Water
Equipment	Centrifugal pump
Feed flow rate (base case)	10,000 kg/h
Inlet temperature (base case)	30 °C
Inlet pressure (base case)	1 bar
Outlet pressure (base case)	5 bar
Specified adiabatic efficiency (base case)	70%
Simulation approach	Steady-state, one-variable-at-a-time sensitivity analysis

4. Process Model Development

A water stream was created and connected to a centrifugal pump. An energy stream was connected to the pump so that HYSYS could calculate the required pump energy duty. The feed conditions were specified at the pump inlet and the required discharge pressure was specified at the outlet. The base case was solved successfully before performing the sensitivity studies.

Flow sheet of Pump with Showing Base Cases

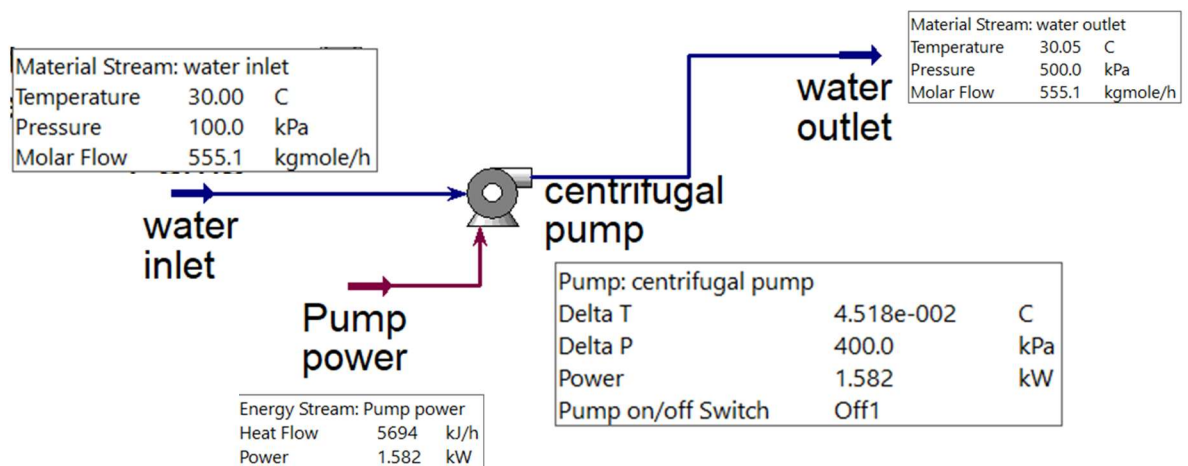


Figure 4.1 Flow sheet of Pump Design

5. Base Case

Parameter	Base-case value
Fluid	Water
Mass flow rate	10,000 kg/h
Inlet temperature	30 °C
Inlet pressure	1 bar
Outlet pressure	5 bar
Pressure rise	4 bar
Pump head	≈ 40.64 m
Specified adiabatic efficiency	70%
HYSYS total/rotor power	≈ 1.582 kW
Outlet temperature	≈ 30.05 °C

The base case provides the reference operating condition for the four sensitivity studies. The pressure rise is calculated as $\Delta P = P_{out} - P_{in} = 5 - 1 = 4$ bar.

6. Methodology

A one-variable-at-a-time approach was used. In each study, one operating variable was changed while the remaining specified conditions were held constant. This allows the effect of each variable to be identified clearly without combining multiple changes in the same test.

6.1 Temperature Study (T1–T5)

Temperature was varied from 20 to 60 °C while flow rate, pressure conditions, and specified efficiency were held constant. As shown in table 6.1

Table 6.1 Showing that Changing temperature how it effects the other conditions ,performance and power of the pump.

case	Changing variable inlet temperature	Pressure head [m]	Total Power [kW]	Total Rotor [kW]	Fluid [kW]	Outlet temperature C	Outlet pressure Kpa	Mass flow kg/hr.	Energy stream heat flow kJ/h	Adiabatic efficiency %
T1	20C	40.34	1.57	1.57	1.57	20.04	500	10000	5652	70
T2	30C	40.64	1.5842	1.5842	1.584	30.05	500	10000	5693	70
T3	40C	40.95	1.5937	1.5937	1.593	40.05	500	10000	5737.48	70
T4	50C	41.27	1.6061	1.6061	1.606	50.05	500	10000	5782.26	70
T5	60C	41.6	1.6189	1.6189	1.618	60.05	500	10000	5828.38	70

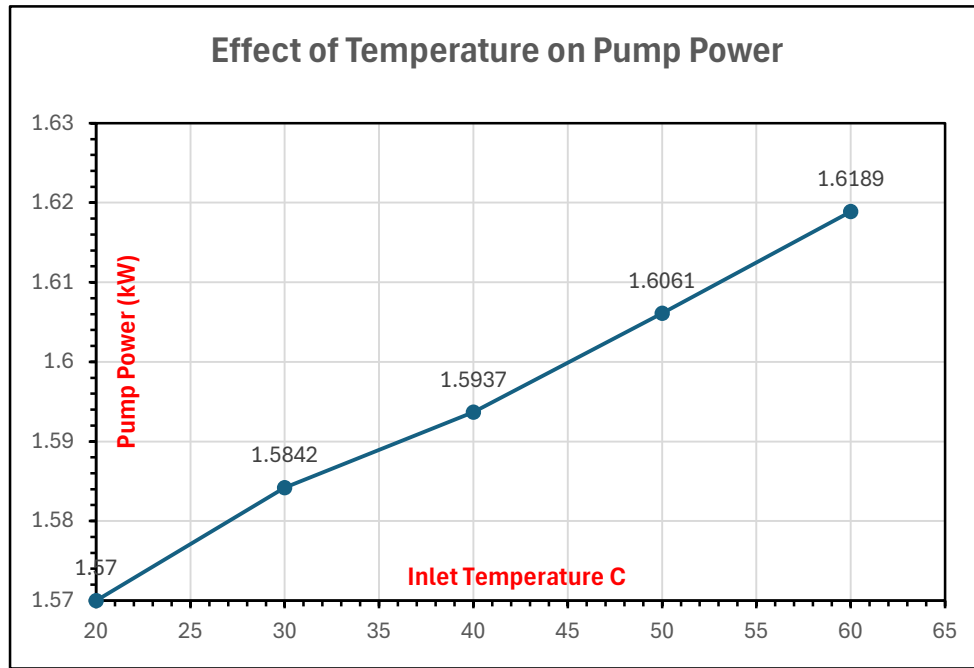


Figure 6.1 : EFFECT OF TEMPERATURE ON PUMP POWER

6.2 Flow-Rate Study (F1–F6)

Flow rate was varied from 5,000 to 20,000 kg/h while temperature, inlet pressure, outlet pressure, and specified efficiency were held constant. As shown in table 6.2

Table 6.2 Show that changing Flow rate of pump how affects the other base conditions , pump performance and pump power

case	Changing flow rate	Pressure head [m]	Total Power [kW]	Total Rotor [kW]	Fluid [kW]	Outlet temperature in C	Outlet pressure in Kpa	Adiabatic efficiency %	Energy stream heat flow kJ/h	Molar Flow Kmol/h
F1	5000	40.64	0.7908	0.7908	0.7908	30.05	500	70	2846.9	277
F2	7500	40.64	1.1862	1.1862	1.1862	30.05	500	70	4270.4	416.4
F3	10000	40.64	1.5842	1.5842	1.584	30.05	500	70	5693	555.1
F4	12500	40.64	1.9771	1.9771	1.9771	30.05	500	70	7117	693.8
F5	15000	40.64	2.3724	2.3724	2.3724	30.05	500	70	8540.9	832.6
F6	20000	40.64	3.1633	3.1633	3.1633	30.05	500	70	11387	1110

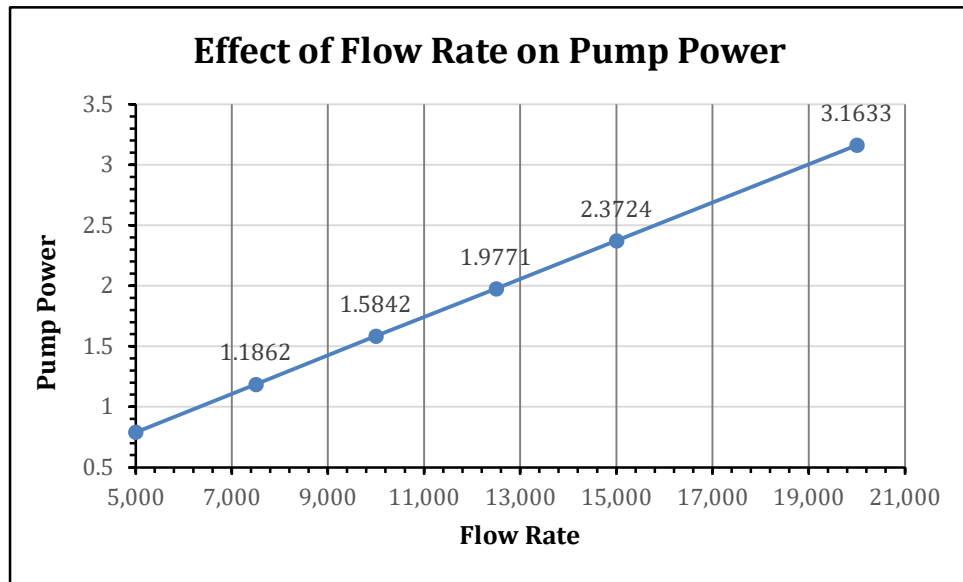


Figure 6.2 This graph shows that increasing the flow rate increase the power consumption as more flow rate power is required more.

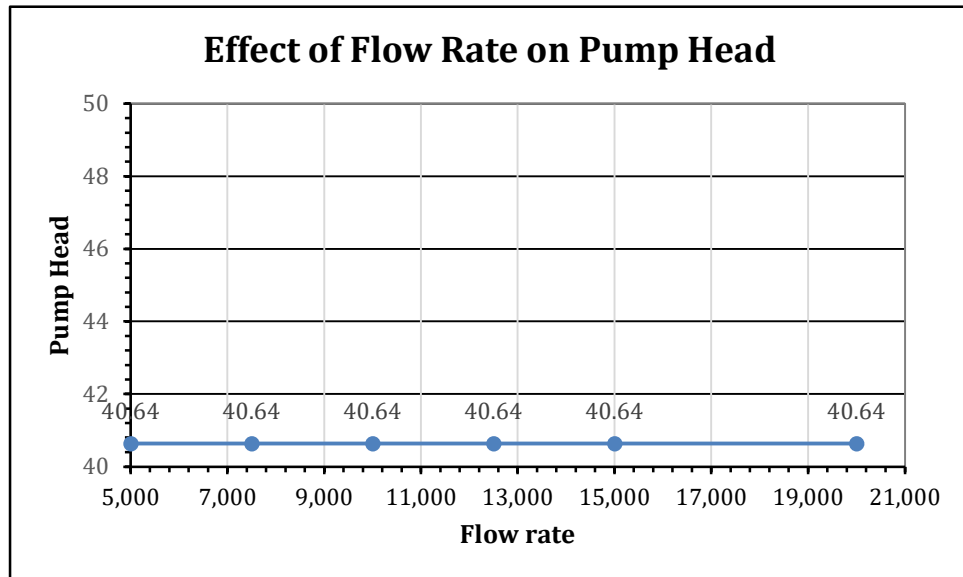


Figure 6.3 Shows that changing Flow rate does not affects the Head of the Pump

6.3 Discharge-Pressure Study (P1–P5)

Outlet pressure was varied from 3 to 7 bar while flow rate, temperature, inlet pressure, and specified efficiency were held constant. As shown in table 6.3

Table 6.3 show that changing discharge pressure how does affects the other base conditions and performance

case	Changing outlet pressure in Bar	Pressure head [m]	Total Power [kW]	Outlet temperature C	Molar Flow Kmole/h	Energy stream heat flow kJ/h	Adiabatic efficiency %
P1	3 bar	20.32	0.7908	30.03	555.1	2846.98	70
P2	4 bar	30.48	1.1862	30.03	555.1	4270.48	70
P3	5 bar	40.64	1.582	30.05	555.1	5693.97	70
P4	6 bar	50.81	1.977	30.06	555.1	7117.47	70
P5	7 bar	60.96	2.3724	30.07	555.1	8540.96	70

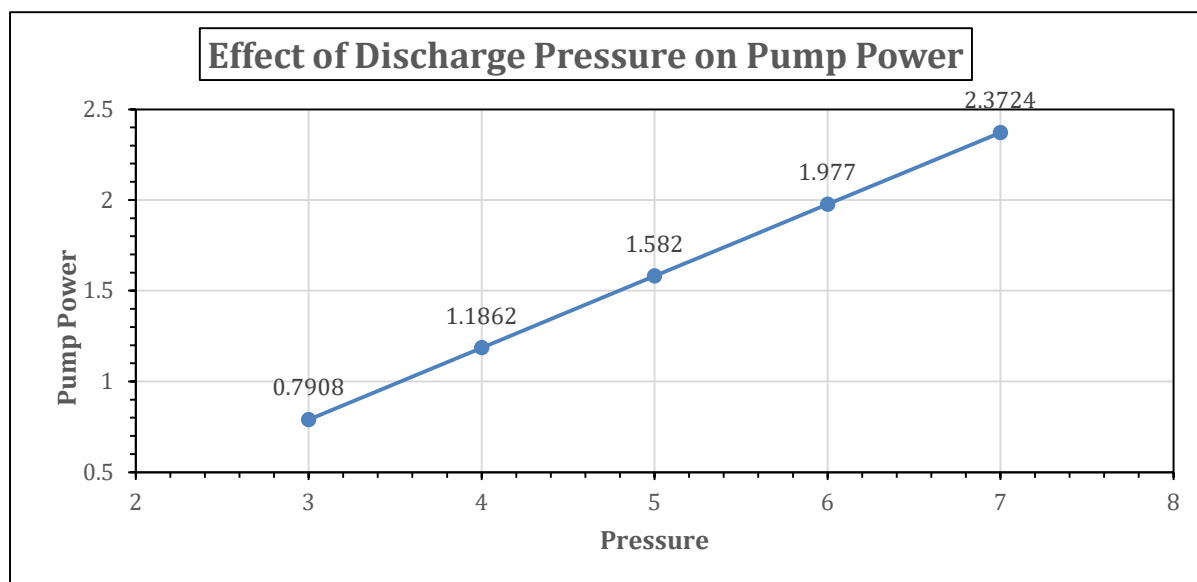


Figure 6.4 : This Graph shows the data base relationship of the Pump power and discharge Pressure that how outlet pressure affects the power of the pump.

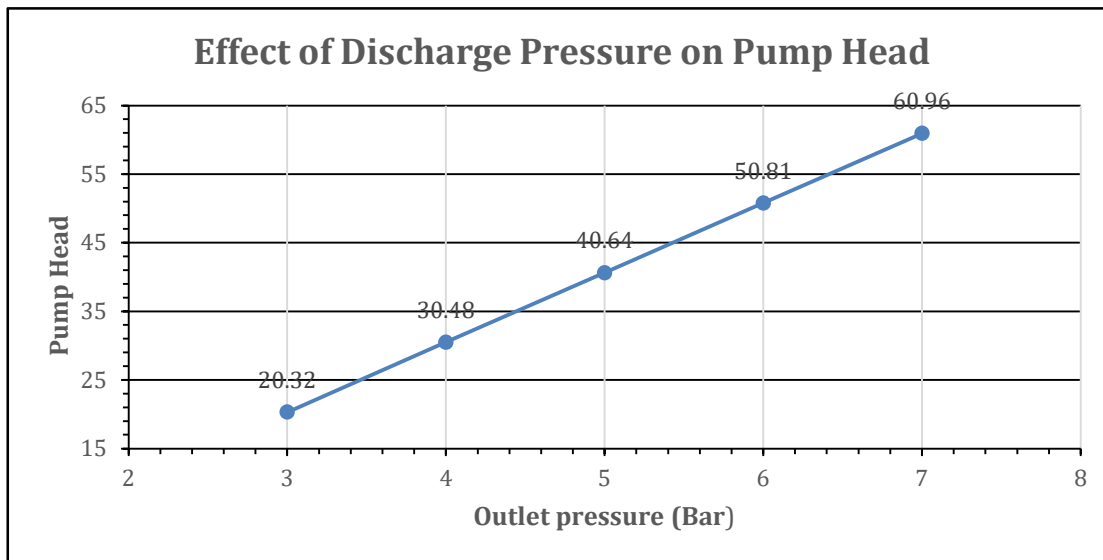


Figure 6.5 : Graph shows the relationship of Pump head and Discharge flow how outlet pressure affects the head of the pump where we can show that increase in outlet temperature increase the head of the pump

6.4 Pump-Efficiency Study (E1–E5)

Specified adiabatic efficiency was varied from 50 to 90% while the hydraulic duty was held constant.as shown in Table 6.4

Table 6.4: show the Changing Pump Adiabatic Efficiency how affects the other Pump performing variables

Efficiency %	case	Pressure head [m]	Total Power [kW]	Outlet temperature C	Outlet P in Kpa	Energy stream heat flow kJ/h
50	E1	40.64	2.2143	30.05	5 bar	7972
60	E2	40.64	1.8452	30.07	5 bar	6643
70	E3	40.64	1.582	30.05	5 bar	5694
80	E4	40.64	1.3839	30.03	5 bar	4982
90	E5	40.64	1.2301	30.02	5 bar	4429

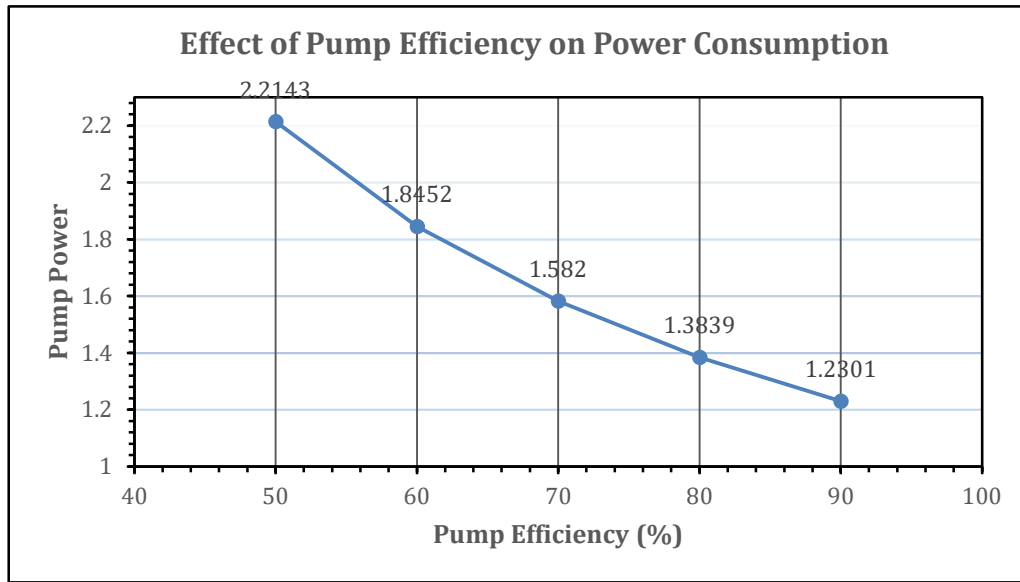


Figure 6.3 : Graph shows the Effects the Pump efficiency on Power Consumption Increasing efficiency decrease the power consumption.

7. Governing Calculations

The main calculations used to interpret the simulation [2,5] results were:

Pressure rise: $\Delta P = P_{out} - P_{in}$

Hydraulic power: $P_{hyd} = \Delta P \times Q$

Pump/shaft power relationship: $P_{shaft} \approx P_{hyd} / \eta$

Specific energy consumption: $SEC = \text{Pump power} / \text{Flow rate (tonnes/h)}$

For the base case, 10,000 kg/h corresponds to approximately 10 tonnes/h. With a 4 bar pressure rise, the hydraulic power is approximately 1.11 kW, while the shaft power expected at 70% efficiency is approximately 1.59 kW. This is consistent with the HYSYS pump power of about 1.582 kW. The distinction between hydraulic power and shaft/rotor power should be maintained when reporting the results.

8. Specific Energy Consumption

Specific energy consumption was calculated as pump power divided by the water throughput in tonnes per hour. For the flow-rate study, the value remained close to 0.158 kWh/tonne because pressure rise and specified efficiency were kept constant. As results shown in table 8.1

Table 8.1 Shows the Pump Specific energy Consumption.

Efficiency	Power (kW)	Flow (t/h)	Specific Energy (kWh/tonne)
50%	2.2143	10	0.22143
60%	1.8452	10	0.18452
70%	1.5820	10	0.15820
80%	1.3839	10	0.13839
90%	1.2301	10	0.12301

Table 8.1 Increasing the specified efficiency from 50% to 90% reduced the calculated specific energy consumption from 0.22143 to 0.12301 kWh/tonne, a reduction of approximately 44.4%.

9. Results and Discussion

9.1 Effect of Flow Rate

Pump power increased from 0.7908 kW at 5,000 kg/h to 3.1633 kW at 20,000 kg/h. The increase is approximately proportional because the pressure rise and specified efficiency were fixed. Pump head remained approximately 40.64 m because the inlet and outlet pressures were fixed at 1 and 5 bar.

9.2 Effect of Discharge Pressure

Increasing outlet pressure from 3 to 7 bar increased pump head from 20.32 to 60.96 m and increased pump power from 0.7908 to 2.3724 kW. This demonstrates the strong effect of pressure rise on pumping energy demand.

9.3 Effect of Temperature

Increasing water temperature from 20 to 60 °C increased calculated pump power from 1.5700 to 1.6189 kW. The change was relatively small compared with the effects observed for flow rate and discharge pressure. The trend is associated with changes in fluid properties such as density and viscosity.

9.4 Effect of Pump Efficiency

Increasing specified efficiency from 50 to 90% reduced pump power from 2.2143 to 1.2301 kW. This corresponds to an approximate 44.4% reduction in required power for the same hydraulic duty. The result demonstrates the importance of pump efficiency in reducing energy consumption.

10. Overall Findings

- Higher flow rate increases pump power consumption under a fixed pressure rise.
- Higher discharge pressure increases both pump head and power consumption.
- Temperature produced a comparatively small change in power over the investigated 20–60 °C range.
- Higher specified pump efficiency substantially reduces required pump power.
- Specific energy consumption decreases as pump efficiency increases.
- For the selected fixed-pressure flow study, pump head remains approximately constant while power varies strongly with flow rate.

11. Engineering Recommendations

- Operate the pump near a high-efficiency operating region whenever process requirements allow.
- Avoid unnecessarily high discharge pressure because excess pressure increases head and energy consumption.
- Avoid unnecessarily high flow rates when the additional throughput is not required by the process.
- Account for changes in fluid properties when pumps are operated over a wider temperature range.
- Monitor flow rate, pressure rise, pump power, and specific energy consumption as practical operating-performance indicators.
- For real pump selection, supplement the sensitivity study with manufacturer pump curves, rotational speed, impeller characteristics, system resistance, and NPSH requirements.

12. Limitations and Scope

This project is a steady-state sensitivity and performance study of a centrifugal pump model in Aspen HYSYS 11. The specified pump efficiency is an input parameter in the efficiency sensitivity study; it is not a predicted pump efficiency curve. Similarly, the study does not represent a manufacturer pump curve or a full system hydraulic design. The results are therefore most appropriate for understanding parameter sensitivity and energy-performance relationships under the selected operating conditions.

13. Conclusion

A centrifugal pump performance analysis was successfully carried out using Aspen HYSYS 11 for water under varying operating conditions. A base case and 21 sensitivity cases were evaluated by changing temperature, flow rate, discharge pressure, and specified pump efficiency. The results showed that flow rate and discharge pressure have strong effects on pump power, while temperature has a comparatively smaller effect within the investigated range. Increasing specified pump efficiency from 50% to 90% reduced required pump power by approximately 44.4% and reduced specific energy consumption from 0.22143 to 0.12301 kWh/tonne. Overall, the study demonstrates how operating conditions and pump efficiency influence energy performance and provides a useful simulation-based framework for evaluating pumping-system operation.

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